

Q U A L I T Y A S S U R A N C E •
A U T O M A T I O N T E S T I N G

TEST PLAN DOCUMENT

BlazeDemo Flight Booking Automation Testing Framework

Selenium WebDriver | TestNG | Maven | Jenkins | Docker | CI/CD

Document Name	BlazeDemo Test Plan
Prepared By	Harsh Choudhary
Trainer	Ms. Vaishali Ahire
Project	BlazeDemo Automation Framework
Version	1.0
Date	June 2026
Status	✓ Approved

Test Plan Identifier

BlazeDemo_TP_001

BlazeDemo Application - Test Plan

Introduction

This test plan outlines the testing activities for the BlazeDemo Flight Booking Application. The objective is to verify the proper functioning of key banking and booking features such as user authentication, flight search, flight selection, payment processing, and booking confirmation. The testing process ensures that the application performs as expected and delivers a reliable user experience.

Test Items

- Home Page (Flight Search Engine)
- Flight Search Results (Dynamic Flight Selection)
- Purchase Page (Passenger Details & Payment Form)
- Booking Confirmation Page

Features to be Tested

- Valid and Invalid Login/Registration (Authentication)
- Departure and destination city dropdowns and selection
- Dynamic flight selection from the availability WebTable
- Purchase form submissions (Valid inputs, Empty submissions, Invalid Data Types)
- Booking confirmation validation (Success message, Booking ID, Amount, Data Integrity)

Features NOT to be Tested

The following areas are excluded from the current testing cycle:

- Performance and load testing
- Security testing (Penetration testing)
- Stress and endurance testing
- Mobile responsiveness testing

Approach

The testing process follows an automation-driven approach using an Enterprise-Grade framework. Selenium WebDriver is used for browser automation, TestNG manages test execution, Maven handles project dependencies, Jenkins supports CI/CD integration, Docker provides containerized execution, and Extent Reports provide detailed execution insights. The framework utilizes the Page Object Model (POM) and ThreadLocal memory management for highly scalable, flake-free execution.

Item Pass/Fail Criteria

The test execution will be considered successful when all planned test cases execute correctly and produce expected results.

- Pass Percentage Target: 100% (Including successful capturing of negative defect scenarios).

Suspension Criteria

Testing activities may be suspended under the following conditions:

- Application becomes fully unavailable or unresponsive.
- Critical blocker defects occur (e.g., "419 Page Expired" preventing access).
- Browser driver initialization failures.
- Docker container or Jenkins pipeline failures.

Resumption Criteria

Testing will resume once the identified issues are resolved and the testing environment is restored:

- Environment restored successfully.
- Reported blocker defects fixed.
- Jenkins/Docker services operational.

Test Deliverables

The following deliverables will be generated during the testing process:

- Test Plan Document
- Test Case Documentation
- Automation Framework Codebase
- Extent Reports (HTML Dashboard)
- Jenkins Pipeline Logs
- Bug Reports (Jira Format)
- Test Analysis Report

Environmental Needs

- **Software Requirements:** Java, Selenium WebDriver, TestNG, Maven, Jenkins, Docker.
- **Browser Support:** Google Chrome (Headless & UI Mode).
- **Hardware Requirements:** Windows 10/11, Minimum 8 GB RAM.

Responsibilities

Harsh Choudhary is responsible for planning and designing test cases, developing automation scripts, executing test scenarios, analyzing results, reporting defects, maintaining the automation

framework, and managing Jenkins CI/CD and Docker containerization activities throughout the project lifecycle.

Schedule

- **Activity 1:** Test Planning & Requirement Analysis
- **Activity 2:** Test Design & POM Implementation
- **Activity 3:** Framework Setup (ThreadLocal, Utilities)
- **Activity 4:** Script Development (Positive & Negative Suites)
- **Activity 5:** Test Execution & CI/CD Pipeline Integration
- **Activity 6:** Reporting & Defect Logging

Risks and Contingencies

The following risks may impact the testing process:

- Instability of the BlazeDemo application.
- Known data integrity limitations (hardcoded dummy data on confirmation pages).
- Unexpected server errors during authentication.
- *Contingency:* Appropriate conditional logic and graceful degradation have been implemented in the automation scripts to handle known limitations without failing the execution pipeline.

Tools Used

Selenium WebDriver, Java, TestNG, Maven, Jenkins, Docker, GitHub, Extent Reports, Jira.

Approvals

- **Trainer:** Ms. Vaishali Ahire
- **Prepared By:** Harsh Choudhary
- **Project:** BlazeDemo Automation Framework

Document Version Information

Field	Value
Document Name	BlazeDemo Test Plan
Version	1.0
Prepared By	Harsh Choudhary
Project	BlazeDemo Automation Framework

Field	Value
Status	Approved
Date	June 2026